APEXPLANET SOFTWARE PVT. LTD.

DATA ANALYTICS INTERNSHIP

EXECUTIVE SUMMARY REPORT

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GitHub Repository:
https://github.com/Nitin6262/apexplanet-data-analytics

Executive Summary

This report presents the work completed during the Data Analytics Internship at ApexPlanet Software Pvt. Ltd. The internship focused on developing practical skills in data analytics by completing five industry-oriented tasks using Python, SQL, Power BI, and machine learning techniques.

The internship began with data cleaning and preprocessing, where the Superstore dataset was prepared by handling missing values, removing duplicates, correcting data types, and ensuring data quality. SQL queries were then used to extract meaningful business information from the dataset through filtering, grouping, aggregation, joins, and sorting operations.

Exploratory Data Analysis (EDA) was performed to understand customer purchasing behavior, sales distribution, profitability, and business trends. Various visualizations such as histograms, scatter plots, box plots, heatmaps, pair plots, and interactive Plotly charts were created to identify patterns and relationships within the data.

An interactive business dashboard was developed using Microsoft Power BI. The dashboard included KPIs, monthly sales trends, category-wise sales, regional performance, top-performing products, customer analysis, and interactive slicers for dynamic filtering. This dashboard provided decision-makers with a clear overview of business performance.

Advanced analytics techniques were implemented to gain deeper insights. Statistical analysis included descriptive statistics, hypothesis testing using t-tests, Chi-Square tests, and confidence interval estimation. Time series analysis was performed by converting transactional data into a time series, resampling it into daily, weekly, and monthly intervals, analyzing trend and seasonality, and forecasting future sales using moving averages.

Customer segmentation was carried out using K-Means Clustering. The optimal number of customer groups was identified using the Elbow Method, followed by visualization

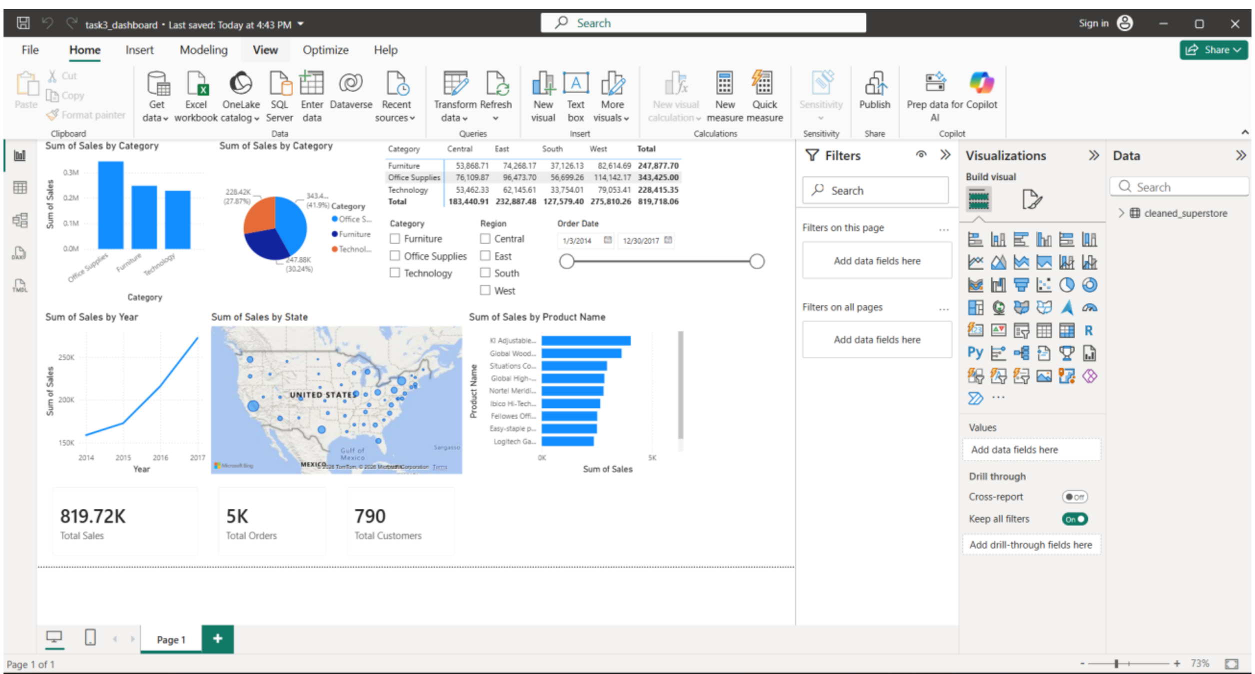
through Principal Component Analysis (PCA). Each customer segment was analyzed to understand purchasing behavior and provide business recommendations.

A Linear Regression model was developed to predict sales using numerical business features. The model was evaluated using R^2 Score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). Feature importance analysis identified the most influential variables affecting sales performance.

Finally, an automation pipeline was created using Python to automatically load datasets, clean the data, generate key performance indicators (KPIs), save processed datasets, and export reports to Excel. This demonstrated how repetitive data processing tasks can be automated efficiently.

Overall, the internship provided valuable practical experience in data cleaning, visualization, business intelligence, statistical analysis, machine learning, and workflow automation while strengthening problem-solving and analytical skills required in the field of Data Analytics.

Dashboard Overview



The Power BI dashboard presents an interactive overview of business performance using multiple visualizations. It includes KPIs for Total Sales, Total Profit, Total Orders, monthly sales trends, category-wise sales distribution, regional performance, and customer insights. Interactive filters allow users to analyze data dynamically based on different business dimensions.

Top Business Insights

1. Technology products generated the highest sales revenue among all product categories.
2. Certain regions consistently outperformed others in both sales and profitability, indicating strong market demand.
3. Customer purchasing behavior varied significantly across different market segments, supporting customer segmentation strategies.
4. Time series analysis revealed noticeable monthly sales trends that can assist in demand forecasting and inventory planning.
5. Machine learning techniques identified important business variables that influence sales performance and customer behavior.

Business Recommendations

• Focus marketing campaigns on high-value customer segments identified through clustering.

• Increase inventory planning for high-demand products based on seasonal sales trends.

• Use predictive analytics to support future sales forecasting and improve business planning.

Technologies Used

- Python
- Pandas
- NumPy
- SQL (SQLite)
- Matplotlib
- Seaborn
- Plotly
- Microsoft Power BI
- Scikit-learn
- SciPy
- Jupyter Notebook
- OpenPyXL

- Git & GitHub

Internship Outcomes

During this internship, practical knowledge was gained in:

- Data Cleaning and Preprocessing
- SQL for Business Analytics
- Exploratory Data Analysis
- Interactive Dashboard Development
- Statistical Analysis
- Customer Segmentation
- Time Series Forecasting
- Machine Learning
- Workflow Automation
- Version Control using Git and GitHub

Conclusion

The ApexPlanet Data Analytics Internship provided hands-on experience in solving real-world business problems using data analytics techniques. Through the successful completion of all five tasks, strong practical skills were developed in data preprocessing, visualization, dashboard development, statistical analysis, machine learning, and automation. The knowledge and experience gained during this internship will serve as a strong foundation for future projects and professional opportunities in the field of Data Analytics and Business Intelligence.